

Progressive web applications

(PWAs): In recent years, PWAs (progressive web applications) have gained substantial attention and popularity. These web-based applications offer users an experience similar to native apps, right within their web browser. PWAs come with several benefits, including offline functionality, push notifications, and quicker loading times. They eliminate the requirement for users to download and install traditional mobile apps while ensuring a smooth and seamless user experience.

Voice user interfaces (VUIs): The emergence of voice assistants such as Amazon Alexa and Google Assistant has led to the integration of voice user interfaces (VUIs) into web applications. VUIs allow users to interact with web applications through voice commands. This functionality proves particularly valuable for hands-free interactions, making it well-suited for smart speakers, voice-enabled devices, and even in-car applications.

Internet of Things (IoT)

connectivity: The utilisation of web applications for controlling and interacting with IoT devices is on the rise. IoT-enabled web applications have the capability to communicate with various devices including smart home appliances, wearables, and industrial sensors. These applications serve as a centralised interface through which users can remotely monitor and control their IoT devices.

Augmented reality (AR) and

virtual reality (VR): Web applications are integrating AR (augmented reality) and VR (virtual reality) technologies to deliver captivating and immersive experiences. AR-enhanced web applications superimpose digital information onto the real world, enriching the user's perception and interaction with their surroundings. On the other hand, VR-enabled web applications create virtual environments, transporting users to entirely new immersive worlds. These

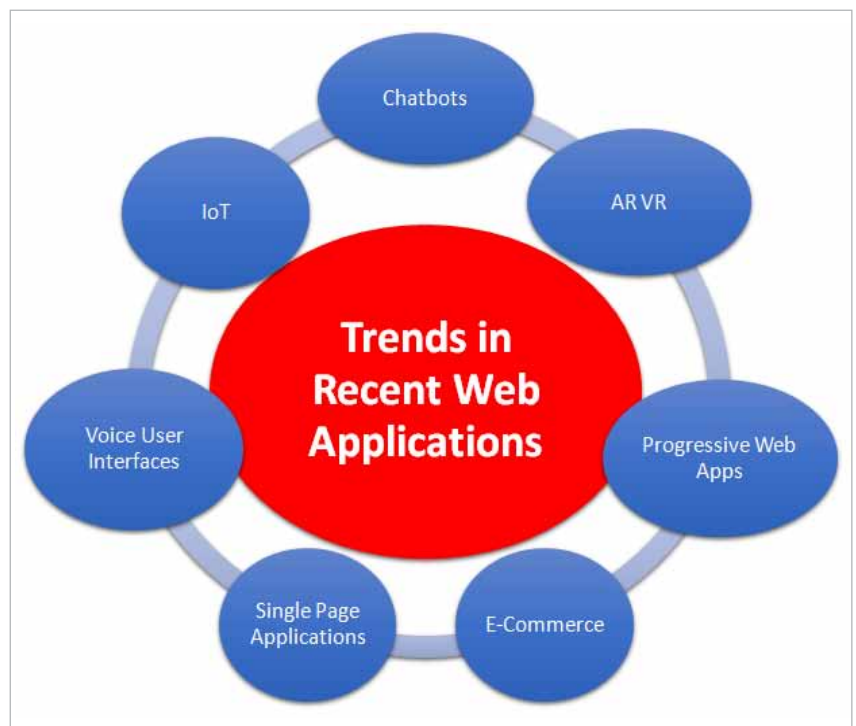


Figure 1: Trends in recent web applications

technologies are being applied across diverse industries such as gaming, e-commerce, real estate, and training simulations, revolutionising the way users engage with digital content.

Single-page applications (SPAs):

SPAs have gained popularity due to their responsive and interactive nature. SPAs load the entire application on a single web page, reducing the need for frequent page reloads. This results in a smoother user experience, faster navigation, and improved performance. SPAs are commonly used for social media platforms, project management tools, and real-time collaboration applications.

E-commerce and online

marketplaces: The expansion of e-commerce has given rise to the emergence of web applications that cater specifically to online retail and marketplaces. These applications serve as platforms for seamless online transactions, efficient product catalogue management, secure payment processing, and personalised

recommendations. Additionally, they incorporate functionalities such as inventory management, order tracking, and customer support systems to ensure a comprehensive and satisfying shopping experience for users.

Collaborative tools and

productivity suites: Web applications are being used to enhance collaboration and productivity in various domains. Applications like project management tools, document collaboration platforms, and video conferencing systems enable remote teams to work together efficiently. These tools often incorporate real-time editing, task assignment, file sharing, and communication features.

Data analytics and visualisation:

Web applications play a vital role in analysing and presenting data in a user-friendly and comprehensible manner. These applications collect, process, and visualise data, enabling users to gain insights and make informed decisions. They find widespread application in areas such as business intelligence, data